CONCORD

Built on the Zama FHE protocol

CONCORD

Trade without showing your hand.

Ethereum Sepolia · euint64 sealed orders · EIP-712 user-decryption

github.com/vihaan1016/zama

THE PROBLEM WITH *open order books*

ON A TRANSPARENT DEX

FRONT-RUN

The MEV tax on every fill

Your order sits in the mempool before it executes — long enough for a searcher to trade ahead of it.

01 **Visible mempool**

Every pending order is public before it settles. Searchers read it, front-run it, and sandwich it.

02 **Limit prices leak intent**

An exposed limit price is a signal. It tells the market exactly where you value the asset before you fill.

03 **Naive batching isn't enough**

Plain batch auctions fix timing, but if the book is still *readable*, the order flow is still exploitable.

ONE SEALED BOOK, *one price*

If no one can read your order before it clears, there is nothing to front-run. The contract discovers the clearing price under FHE — without ever seeing an individual bid.

01 Sealed submission

Price and size are FHE ciphertexts before they touch the mempool. Only the side is public — the market sees a slot fill.

02 Uniform price under FHE

On close the contract computes one clearing price over the encrypted book — everyone crosses at the common price.

03 Confidential settlement

Crossing orders settle as ERC-7984 transfers. You decrypt only your own fill — no one reads anyone else's.

04 32-tick price grid

Limit prices snap to a discrete on-chain grid ($0.01 → $3.11), keeping the encrypted math inside euint64.

encrypted client-side • keeper clears the sealed book • you decrypt only your own fill

BATCH LIFECYCLE

FIVE STATES, *keeper-driven*

A batch opens for a fixed window. A keeper drives every transition — closing the batch, scanning the encrypted grid in gas-sized chunks, public-decrypting the winner, and settling confidentially.

`Open` → `Closed` → `Clearing` → `Cleared` → `Settled`

01
Open → Closed

Encrypted orders accumulate for the batch window; at the deadline the keeper closes submission.

02
Clearing

The contract scans the 32-tick encrypted price grid in paginated, gas-sized chunks under FHE.

03
Cleared

The winning tick is public-decrypted — the one uniform price plus the matched volume, an aggregate only.

04
Settled

Crossing orders move as confidential ERC-7984 transfers. Each trader decrypts their own fill.

LIVE DEMO

END-TO-END *on Sepolia*

Connect a wallet and run the full loop — six steps, no leaving the app.

01 **Claim from the faucet**

Mint confidential test tokens to your wallet in one click.

02 **Approve the DEX operator**

Authorize the DEX to move your ERC-7984 balances.

03 **Encrypt & submit**

Pick a side + tick + size; encrypt client-side; submit the sealed order.

04 **Batch closes**

Watch the sealed book fill — slots, never contents — then the window shuts.

05 **Keeper clears**

One uniform price is revealed over the encrypted book.

06 **Decrypt your fill**

EIP-712 user-decryption reveals only your own result.

COVERAGE

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ENTER *the app*

Trade without showing your hand. The landing lays out the whole thesis: sealed submission, one uniform price, confidential settlement.

connect → trade → view batches

A cream "ledger-paper" interface — every number in mono, nothing about your order ever rendered to anyone else.

CORRECTNESS

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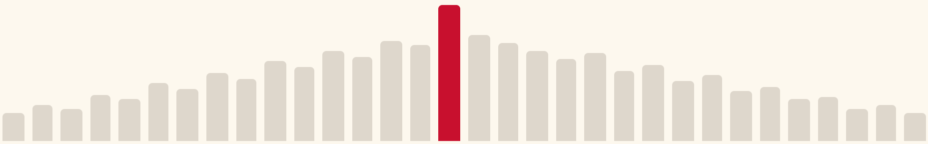
SEAL *your order*

BUY

SELL

LIMIT PRICE

$1.51 · TICK 15



$0.01

$3.11

SIZE

250

Pick side, tick, and size. The limit price snaps to a discrete 32-tick grid — the same `TickMath` the contract runs.

side • tick • size → encrypt

The tick index (0—31) and the size are both FHE-encrypted in the browser as `euint64` ciphertexts. Only the side stays public.

CONFIDENTIALITY

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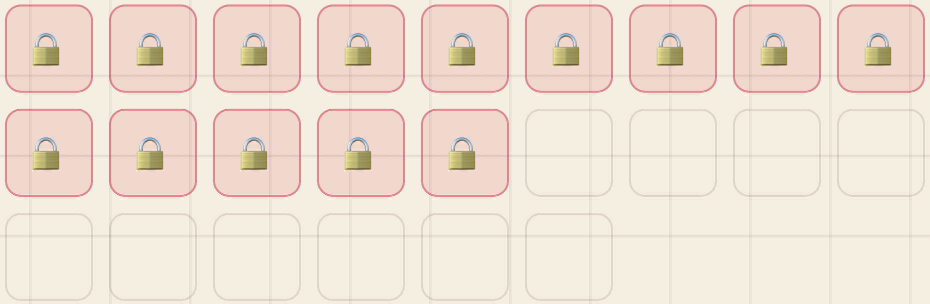
THE SEALED *book*

The market watches slots fill — never their contents. Each locked cell is one order in the batch. Price and size stay encrypted on-chain.

order in → 🔒 slot → book grows

This is the whole point: there is no readable order flow to trade against. A searcher sees a count go up, and nothing else.

SEALED ORDERS (14)



CORRECTNESS

CLEAR *at one price*

UNIFORM CLEARING PRICE

$1.51

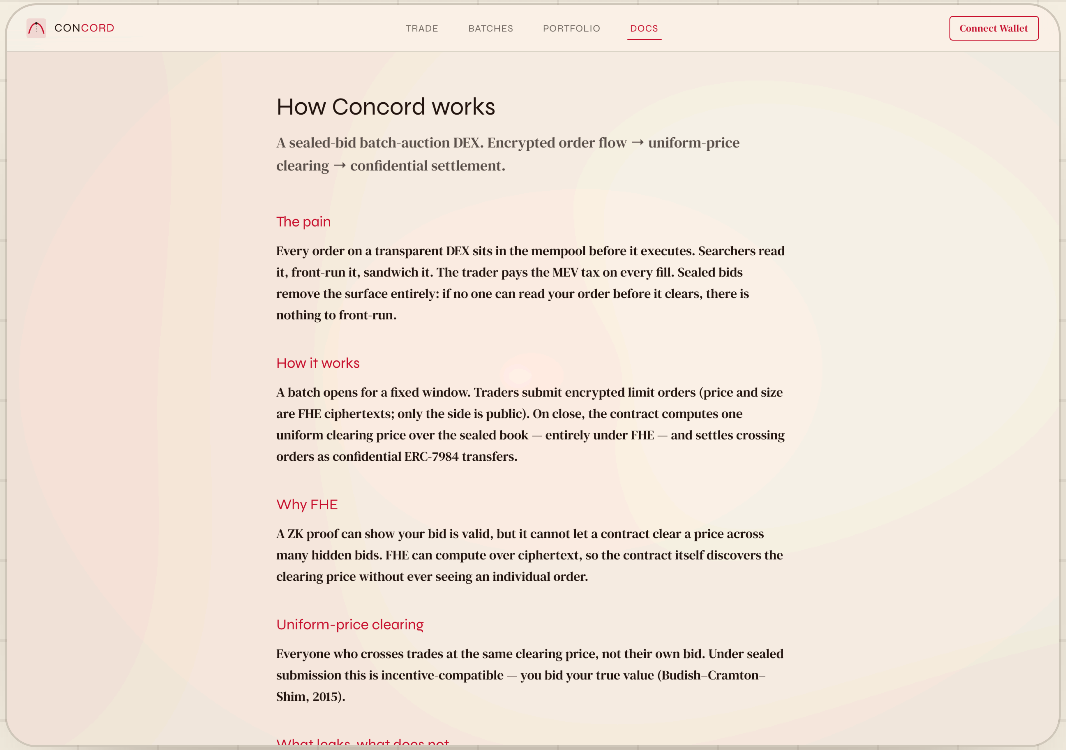
matched volume 1,840

Everyone who crosses trades at the same price — not their own bid. Under sealed submission that's incentive-compatible: you bid your true value.

scan grid → public-decrypt winner

Clearing reveals only the winning tick and the matched volume — an aggregate. Individual orders never leak. *(Budish—Cramton—Shim, 2015.)*

DECRYPT *your fill*



CONCORD

TRADE BATCHES PORTFOLIO DOCS

Connect Wallet

How Concord works

A sealed-bid batch-auction DEX. Encrypted order flow → uniform-price clearing → confidential settlement.

The pain

Every order on a transparent DEX sits in the mempool before it executes. Searchers read it, front-run it, sandwich it. The trader pays the MEV tax on every fill. Sealed bids remove the surface entirely: if no one can read your order before it clears, there is nothing to front-run.

How it works

A batch opens for a fixed window. Traders submit encrypted limit orders (price and size are FHE ciphertexts; only the side is public). On close, the contract computes one uniform clearing price over the sealed book — entirely under FHE — and settles crossing orders as confidential ERC-7984 transfers.

Why FHE

A ZK proof can show your bid is valid, but it cannot let a contract clear a price across many hidden bids. FHE can compute over ciphertext, so the contract itself discovers the clearing price without ever seeing an individual order.

Uniform-price clearing

Everyone who crosses trades at the same clearing price, not their own bid. Under sealed submission this is incentive-compatible — you bid your true value (Budish–Cramton–Shim, 2015).

What leaks, what does not

Your own fill is revealed through the **EIP-712 user-decryption** flow against the relayer.

sign once → session cached → your fill

Decrypting anyone else's fill fails the on-chain access check. The public only ever saw a ciphertext handle.

THE CORE GUARANTEE

YOU CAN'T READ *my order*

WHAT YOU CAN DECRYPT

Your own fill

An EIP-712 request signed by your wallet authorizes decryption of handles the on-chain ACL granted to *you*.

WHAT YOU CANNOT

Anyone else's order

Point the same flow at another trader's handle and it fails the access check. Privacy is enforced on-chain, not by the UI.

FHE + on-chain ACL • the mempool only ever saw an encrypted handle

EXTENSIBILITY

32 TICKS NOW, *parameterized*

The grid is a constant, not a hard-coding. Widening it is a parameter change — the encrypted math and the keeper's chunked scan already generalize.

A • Price grid — `lib/ticks.ts`

```
// widen the grid → more ticks
export const MAX_TICK = 31
export const TICK_SPACING_WEI = 10n ** 17n
export const MIN_PRICE_WEI  = 10n ** 16n
```

B • Indexer — metadata only

```
// never price / size — those stay
// encrypted on-chain
batch: { id, status, orderCount,
         clearingTick, matchedVolume }
order: { batchId, trader, side }
```

read-only indexer • chain → REST + socket • the sealed fields are never persisted off-chain

JUDGED *on*

01 **Confidentiality**

Price and size are euint64 ciphertexts end-to-end; only the side and post-clear aggregates are ever public.

02 **Correctness**

Encrypt → submit → clear → settle → user-decrypt produce the expected on-chain results; 11/11 contract tests pass.

03 **MEV-resistance**

No readable order flow before clearing — nothing to front-run or sandwich, by construction.

04 **UX**

Faucet, operator approval, sealed submit, live batch fill, and clearing reveal — every state animated, never a raw revert.

05 **Code quality**

Typed end-to-end; FHE isolated in hooks; keeper + read-only indexer; metadata-only off-chain store.

06 **Production-readiness**

Sepolia + Zama relayer, public-RPC fallback, keeper in docker-compose, resumable paginated clearing.

CONCORD

Sealed bids, uniform price, confidential settlement.

THANK *you*

github.com/vihaan1016/zama

Not affiliated with Zama. Built on the public Zama Protocol.